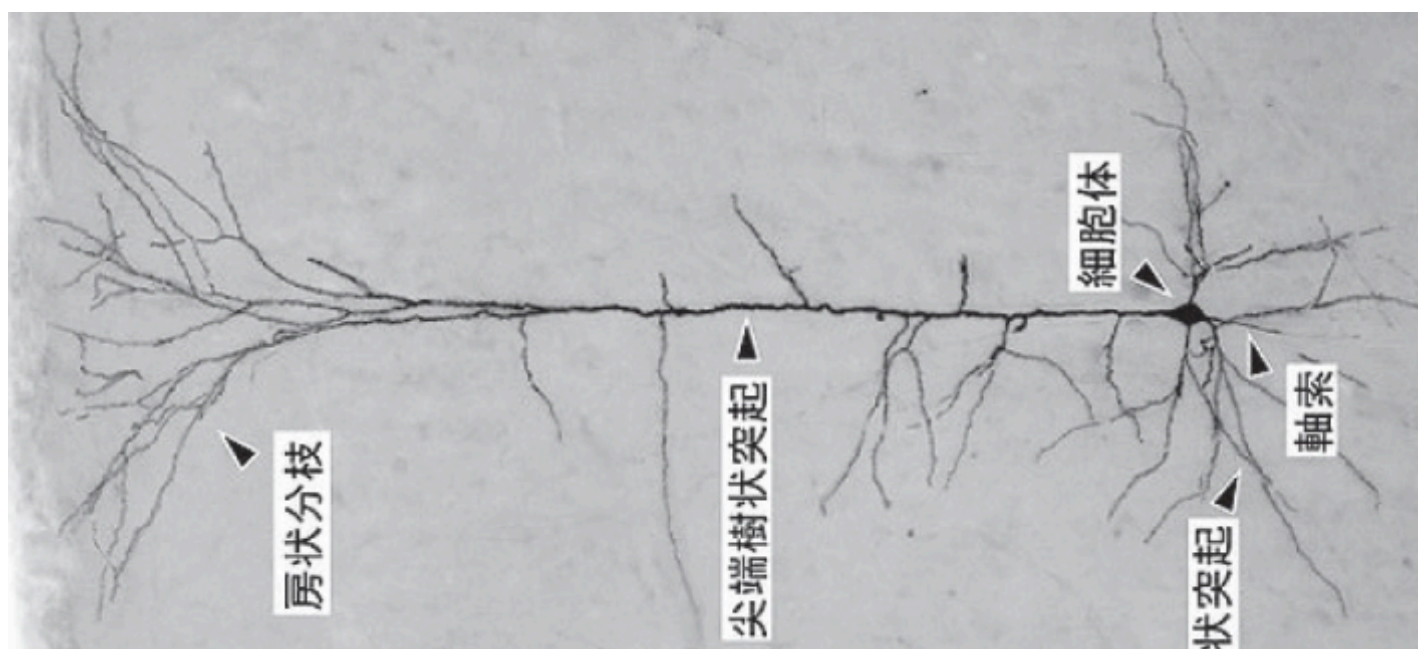


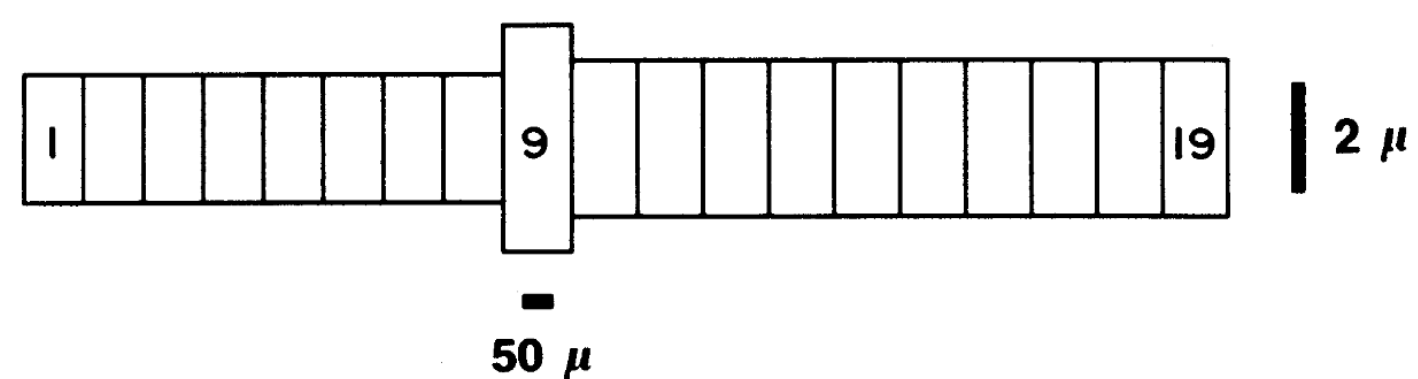
A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

Haruki Munechika

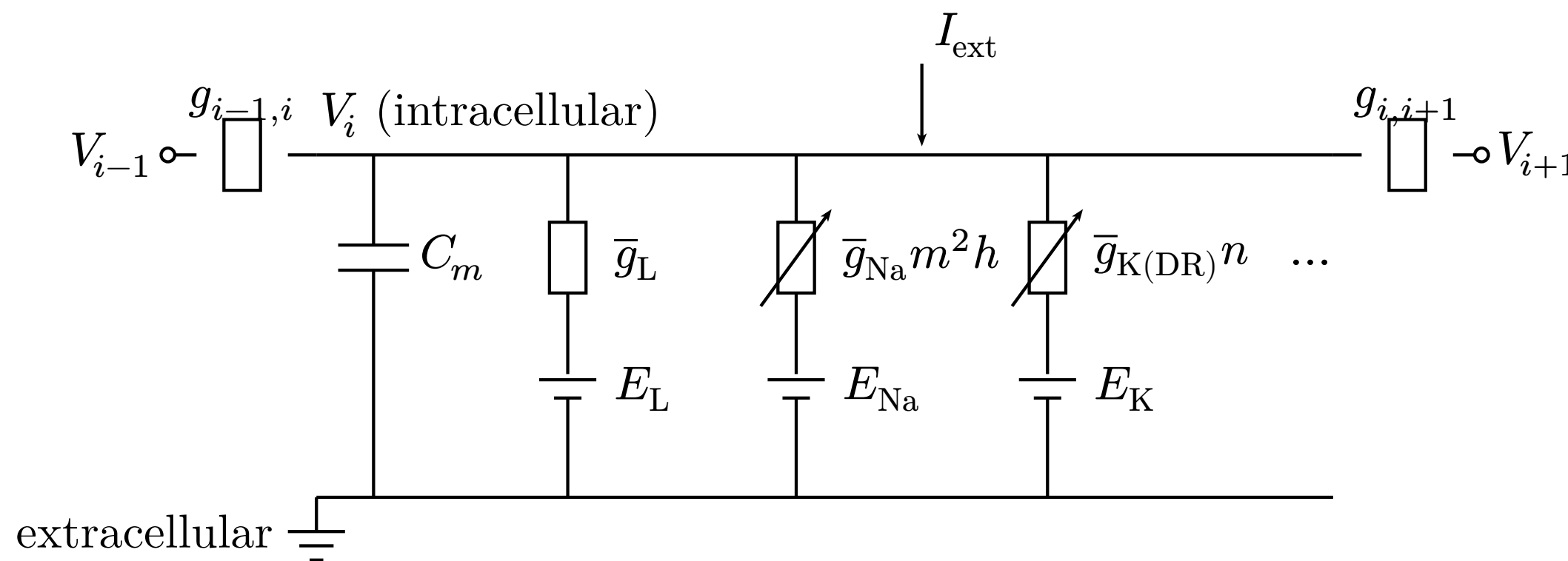
Introduction



CA3 pyramidal neuron [1]



Modelled as 19 compartments [2]



Equivalent circuit of compartment i

Each compartment: capacitance, **variable** ionic conductances, axial coupling $g_{i,i\pm1}$.

Conductances depend on **gate variables**, ODEs with rate functions α, β :

$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (V - E_{\text{Na}})$$

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$

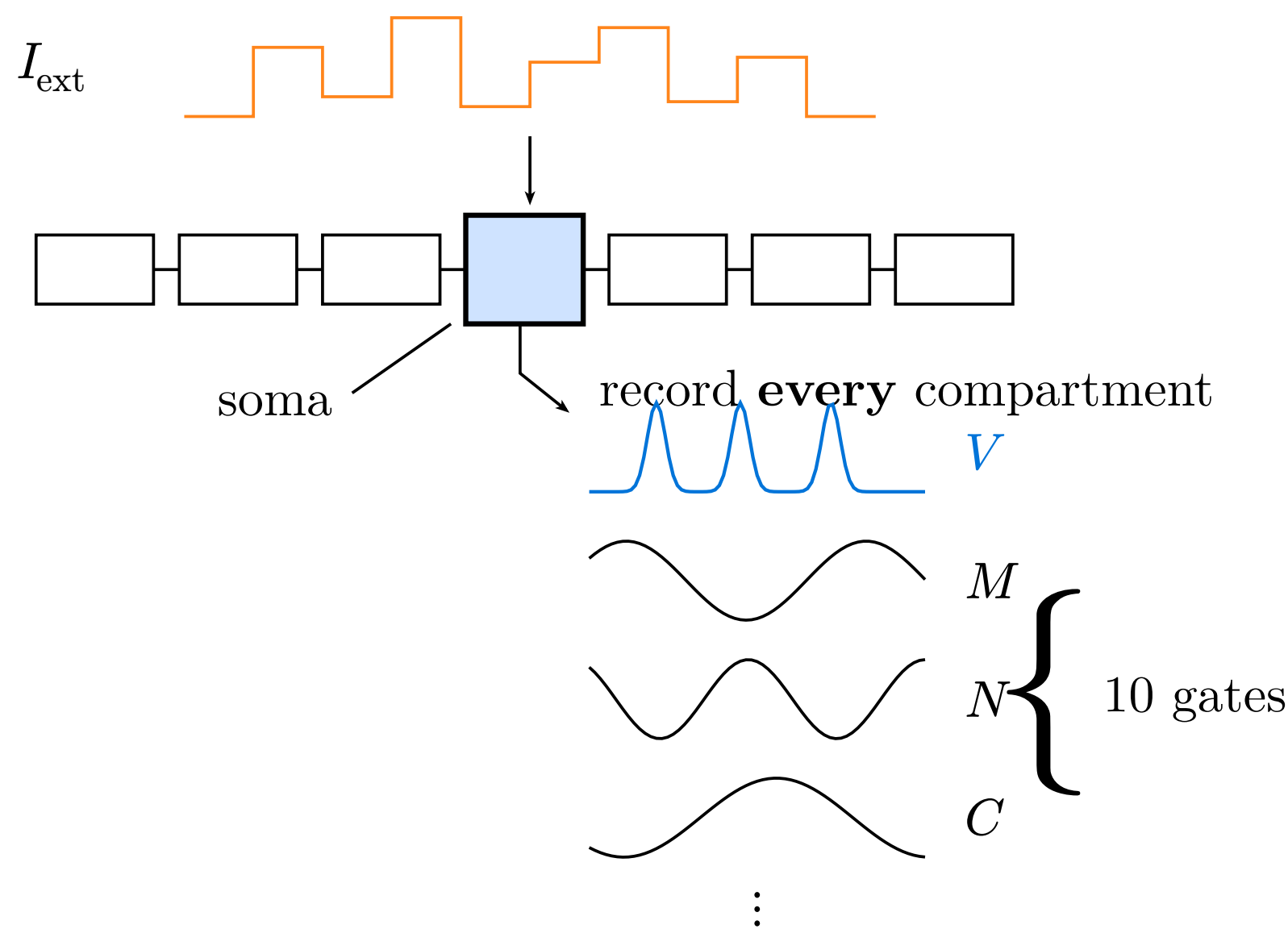
Na current, its gate m , and one rate function ($u = V - E_L$) $\rightarrow$ **11 states per compartment** $\rightarrow$ **209** for 19; at brain scale the gates become a **memory bottleneck**.

Goal

Test whether **complex gate dynamics** remain tractable to a **simple regression** (SINDy) after a coordinate change — and thereby reproduce the membrane-potential response with **fewer state variables**.

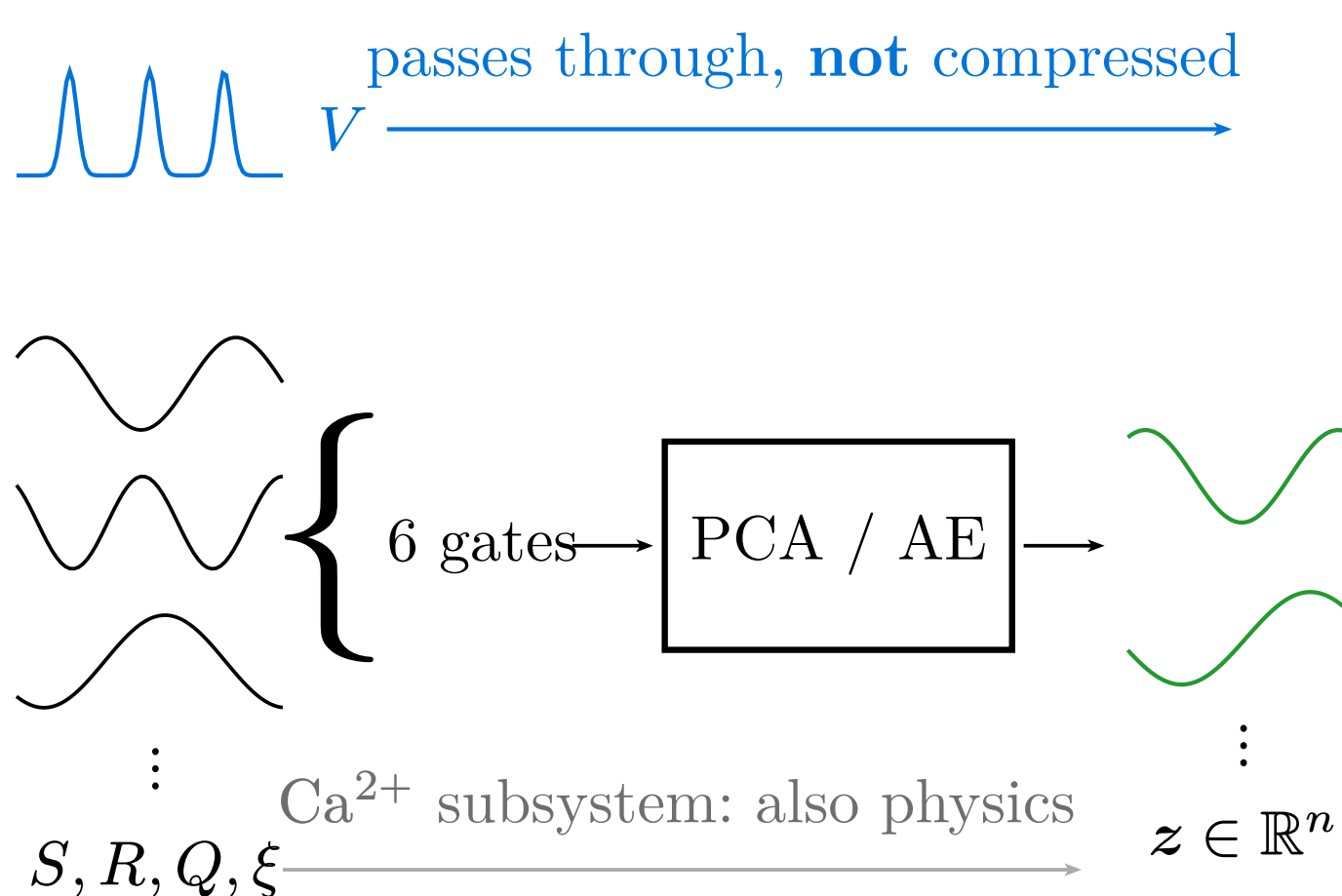
Methods

① Collect the training data



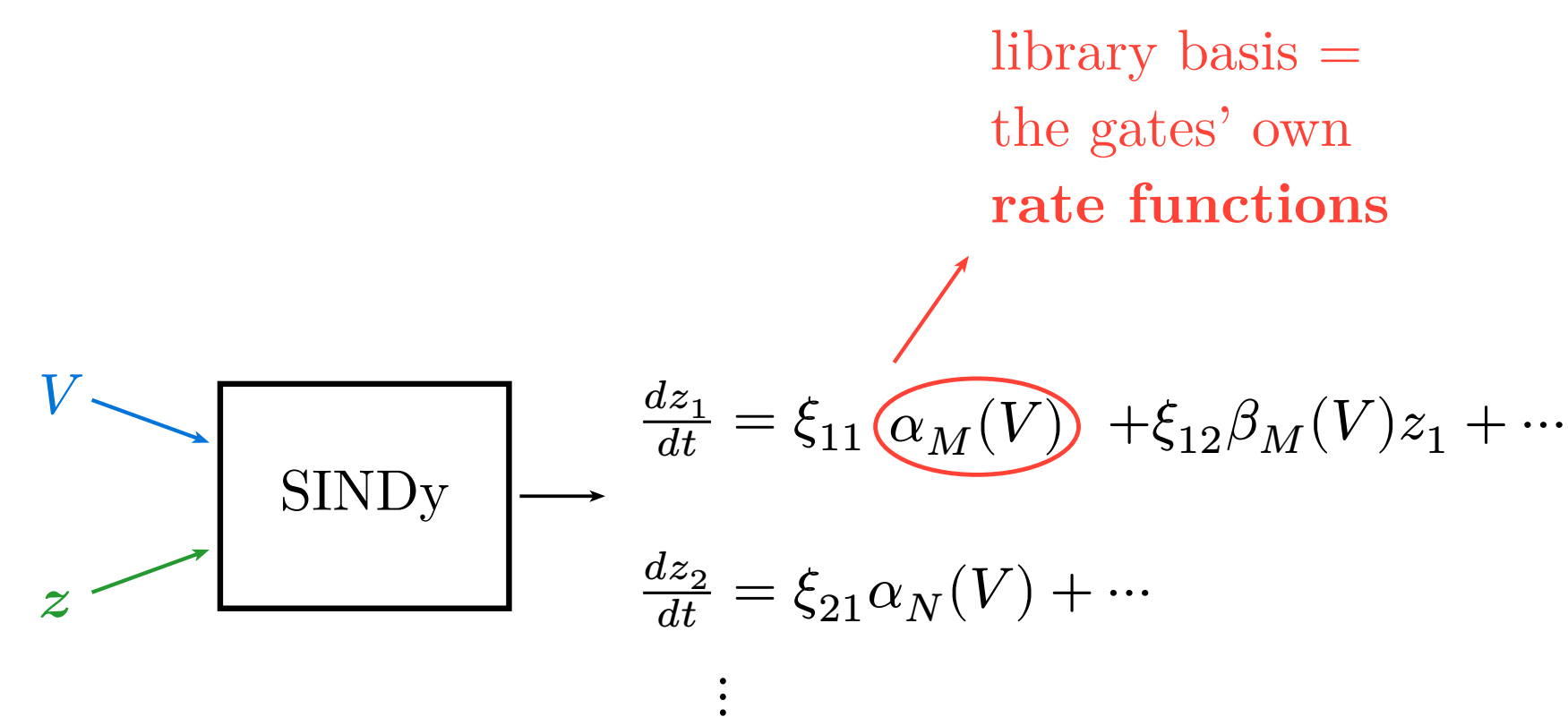
Inject a **random pulse train** at the soma; record V and **10 gates** for all **19 compartments**.

② Compress *only* the gates



6 voltage-dependent gates ride a **low-dimensional manifold** $\rightarrow z \in \mathbb{R}^n$, $n = 5$. V and Ca^{2+} **stay uncompressed**.

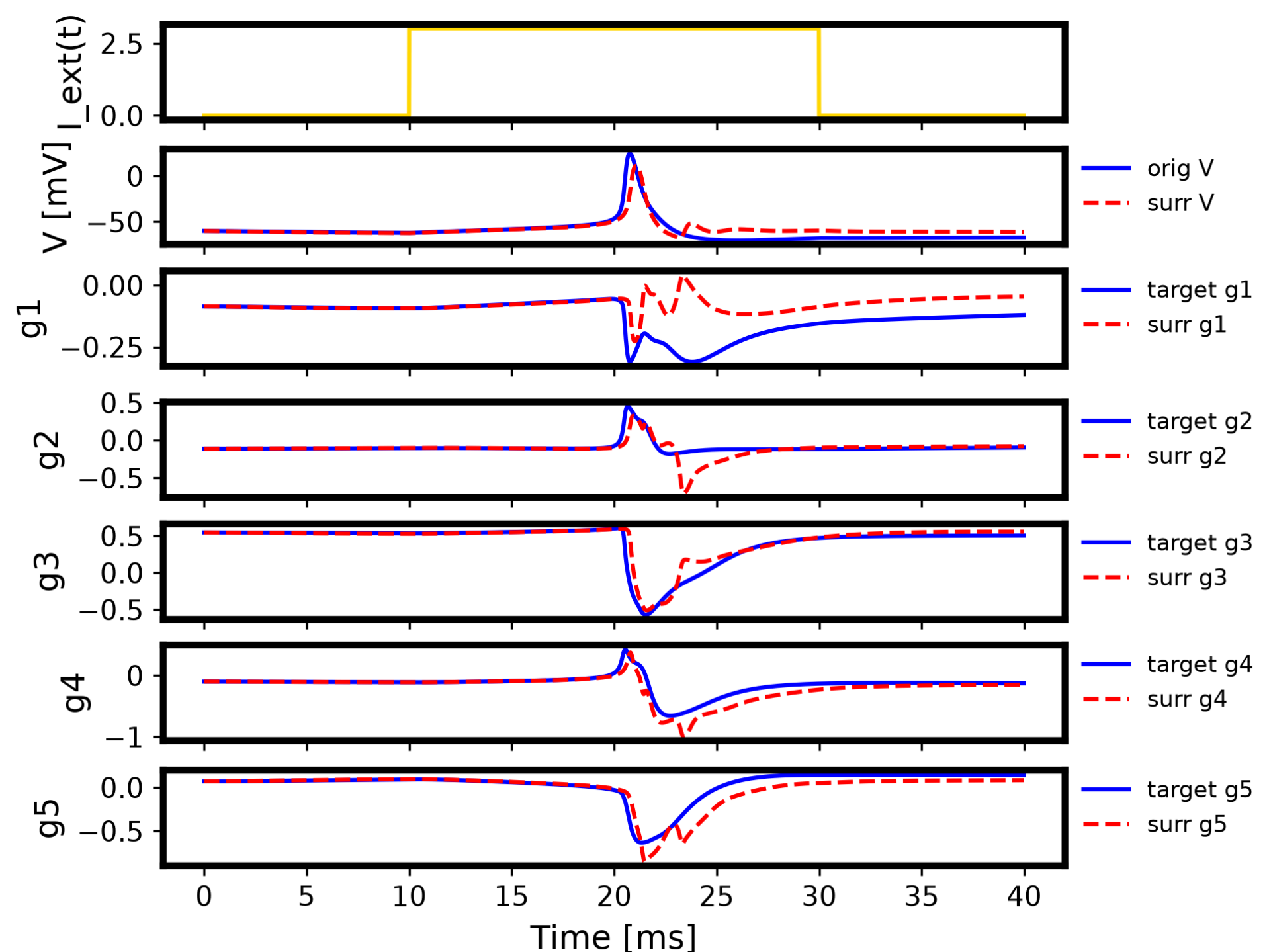
③ Identify the latent ODEs



SINDy [3] fits **only $\dot{z}$** ; $\dot{V}$ and Ca^{2+} keep **original physics**. Library is **physics-informed**: gates' own $\alpha(V), \beta(V)$, not a **41-term** generic one.

Results and Discussion

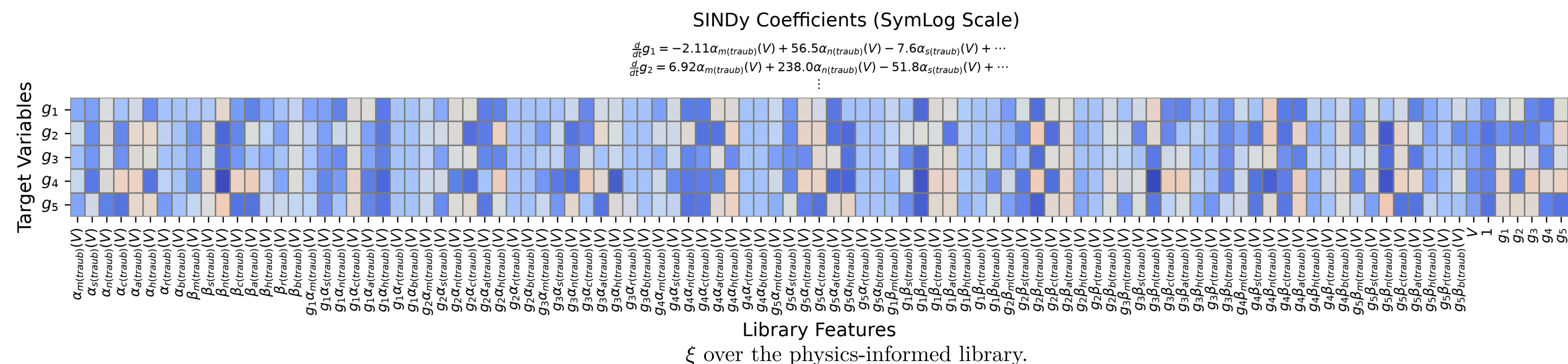
① Single action potential



20 ms, $3 \mu\text{A}/\text{cm}^2$ step: V and the **5 AE latents** (AE, $n = 5$).

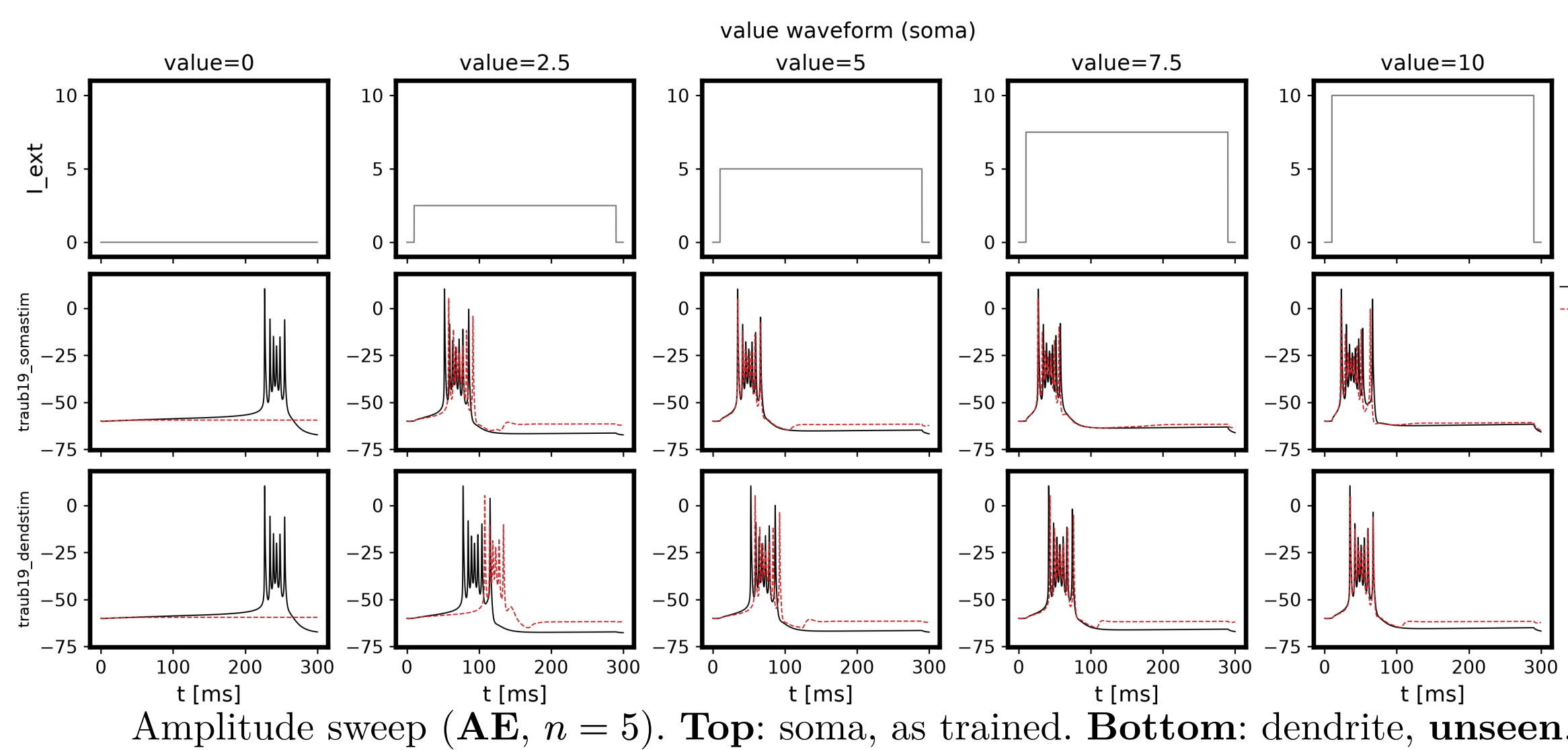
- When it fires is accurate (latency err **0.3 ms**, AHP timing gap **3.1 ms**) — latent dynamics preserve **timing**
 - How big is systematically **underestimated** — peak **13 mV** low (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**: error concentrates in the **fast, steep transition**, not the slow relaxation
- $\rightarrow$ suggests the **AE encoding** ($n=5$), not the SINDy library, fails to preserve the **steepest transition**

④ Identified latent equations



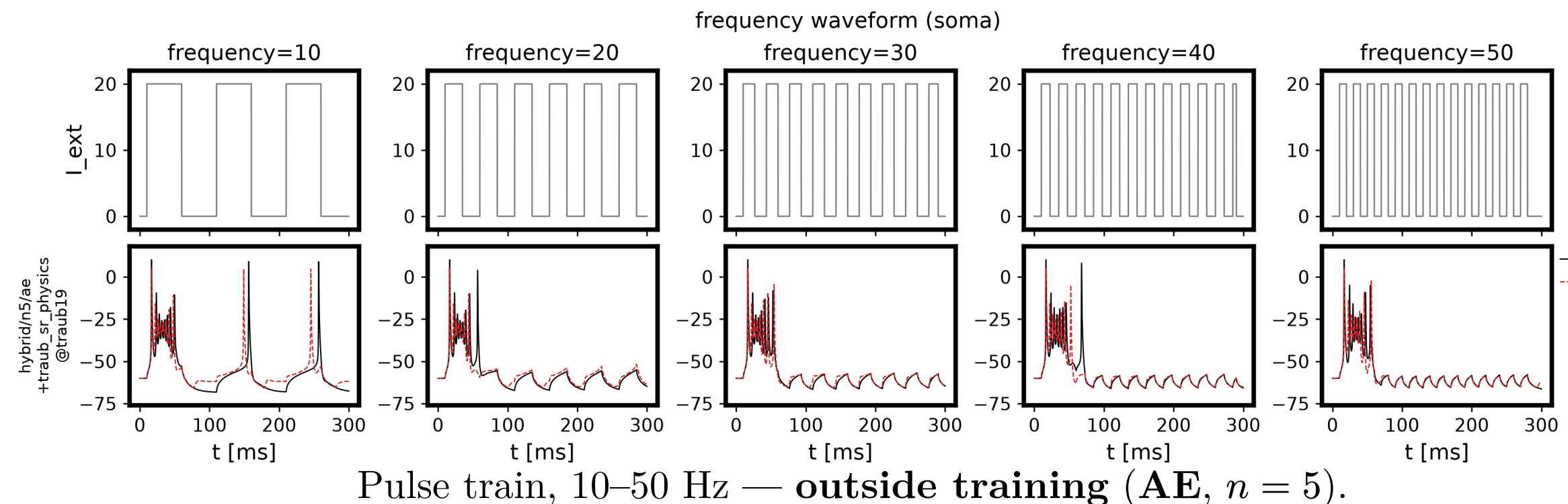
- 79.6% non-zero** — accurate but **not sparse**.

② New stimulus site



- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

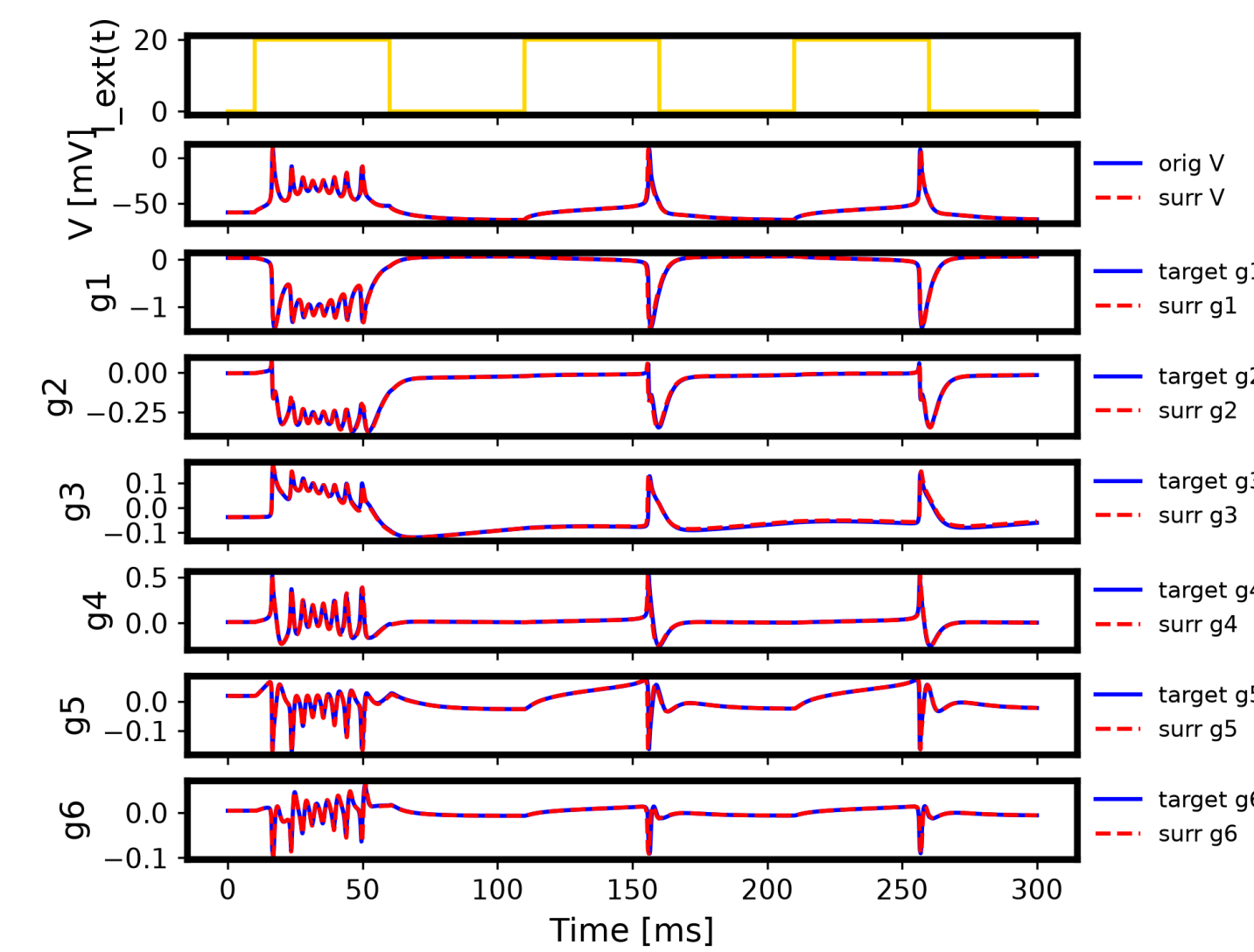
③ Unseen periodic drive



Pulse train, 10–50 Hz — **outside training** (AE, $n = 5$).

- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

⑤ PCA, no compression



PCA, $n = 6$ = uncompressed (vs. AE, $n = 5$ above).

- Holds even **without compression** $\rightarrow$ what matters is the **coordinate change itself**, not the compression ratio: core evidence that complex gate dynamics stay **tractable to simple regression**.

Conclusion

- Replaced **all 19 compartments**, no divergence; transfers to **unseen site** · **frequency**.
 - Core finding**: SINDy succeeds **not because we compressed** ($n: 6 \rightarrow 5$) — ⑤ shows the **same result uncompressed** ($n = 6$), so it is the **coordinate change itself** that makes gate dynamics learnable.
 - Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).
- $\rightarrow$ **Future**: sparsify ξ , push n down, test on other channel models.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

- “錐体細胞.” Accessed: Dec. 08, 2025. [Online]. Available: <https://doi.org/10.14931/bsd.2091>
- R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, “A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances,” *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.
- K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, “Data-Driven Discovery of Coordinates and Governing Equations,” *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.